# What Transfers to a New Benchmark? Priors, Not Item Estimates, for Label-Efficient LLM Evaluation

**Arshia Teymourtash***
KTH Royal Institute of Technology
Stockholm, Sweden
arturt@kth.se

**Ankit Grover***
KTH Royal Institute of Technology
Stockholm, Sweden
agrover@kth.se

## Abstract

Evaluating an LLM on every item of every benchmark is expensive, so predicting item outcomes from a few revealed answers is attractive, but only if what is learned from past benchmarks carries over to new ones. We ask what does, in a controlled comparison where the models, the evaluation tasks and the order in which answers are revealed are held fixed and only the source of the prior changes. A prior learned from past benchmarks transfers to a new benchmark. Estimates of individual item difficulty do not. The prior's benefit is small on average and depends on the benchmark: it helps on benchmarks of typical difficulty and hurts on much easier or much harder ones. A neural network that predicts item difficulty from item text adds little beyond the prior. Whether pooling answers across LLMs helps remains open, because our evaluation reveals answers from one LLM per benchmark. Our entry to the Predictive AI Evaluation Competition (Codabench submission 1005155) is a Beta-Binomial predictor. We release the simulation code, the fixed evaluation samples and every saved score at https://github.com/Agrover112/aims-neurips/tree/submission.

## 1 Introduction

Evaluating language models on every item of every benchmark is expensive, so a predictor that infers a model's per-item results from a handful of labelled answers is attractive. The natural recipe is to learn from past benchmarks how items behave and carry that to a new benchmark. Whether it carries is an empirical question, and the answer matters because new benchmarks and new models keep arriving. Learned predictors are known to fail on unfamiliar models while estimators anchored to the observed labels survive [12], and amortized item-response models learn item difficulty from item text [11]. We ask what, among the things one can fit offline, actually transfers to an unseen benchmark.

We use the Predictive AI Evaluation Competition's setting: a *task* is one subject (a model under given evaluation settings) answering one benchmark, labels are bought one at a time, and predictions are scored by Brier ALC at 0, 1, 3, 7, 15 and 31 labels. Our contributions:

- **A controlled comparison.** We hold the models, the tasks and the order of revealed labels fixed and change only where the prior comes from. We run it in an online simulation of the competition protocol (Section 4).
- **What transfers.** A prior learned from past benchmarks transfers to a new benchmark, but estimates of individual item difficulty do not (Section 5). The prior helps on benchmarks of typical difficulty and hurts on much easier or much harder ones. Neural networks that

*Work done equally.

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predict item difficulty from item text, and priors that adapt to the first labels, add almost nothing beyond it (Section 5.4).

- **Open materials.** The simulation code, the fixed evaluation samples and every saved score are at `https://github.com/Agrover112/aims-neurips/tree/submission`.

## 2 Related work

Empirical Bayes estimates a prior from related observations; classical shrinkage and small-area estimation show how pooling can stabilize noisy group estimates [1–3]. Our Beta-Binomial baseline is its simplest instance: a Beta prior on a task's accuracy, fixed in advance (Jeffreys [9]), tuned on validation, or fitted across training tasks. Prediction-powered inference also combines predictions with observed labels, but targets uncertainty about population quantities rather than sequential item-level prediction [4].

Item response theory explains a response by the subject's ability and the item's difficulty; the Rasch model is its basic form [10]. Extensions model item–respondent interactions in a latent space or separate language effects from item content [5, 6], and amortized IRT predicts item parameters from text [11]. Work on benchmark prediction from fewer data reports that learned predictors fail on unfamiliar models while estimators anchored to the observed labels survive [12]. We ask the analogous question for unfamiliar benchmarks: whether item-level signals and transferred priors remain useful on a new benchmark, where only a few labels are available.

## 3 Data and evaluation protocol

**Data.** The data are a binary response matrix $Y \in \{0,1\}^{N \times M}$ with $N$ subjects (rows, index $i$) and $M$ items (columns, index $j$): $Y_{ij} = 1$ if subject $i$ answers item $j$ correctly. Most entries are missing, since a subject answers only the items of the benchmarks it was evaluated on. Each item $j$ belongs to the dataset $d(j)$ of one of $D = 17$ benchmarks, so a benchmark is a block of columns, and a *task* is one row restricted to one block: subject $i$ on benchmark $d$.

Each item $j$ has an embedding $x_j \in \mathbb{R}^{1024}$ computed once from its text by the frozen BGE-M3 encoder [8] with a context of 8,192 tokens, and normalised to unit length. We use BGE-M3 because the corpus is multilingual (it includes Chinese, Russian and several African languages) and some items are long, which English-only and 512-token encoders handle poorly.

**Online simulation and splits.** The competition protocol is *online*: for each task, labels are revealed one at a time and the predictor updates from them. Whatever the predictor knows before the first label is *offline*, that is, fitted on the training benchmarks beforehand. Our *online simulation* runs this protocol on recorded responses, with disjoint column blocks for training, validation, and test (11, 3, and 3 benchmarks): priors are fitted offline on training, and validation selects settings. The validation and test benchmarks are *out of distribution* (OOD): their responses are excluded from offline fitting.

A *competition-shaped draw* samples one subject per benchmark, so a predictor cannot borrow labels from other subjects on the same benchmark; this simulates the competition protocol on public data. As a separate *in-distribution* (ID) evaluation, we hold out rows (subjects) within the training blocks, so its benchmarks and items have been seen during fitting, and retain the questions covered by the fit subjects. The offline comparator reuses its fitted benchmark rates and item difficulties where available, and its online update changes only the subject ability.

**Budgets and scoring.** For each subject–benchmark pair we freeze a split of distinct items into acquisition and evaluation pools, assigning the extra item to acquisition when the count is odd. A frozen random acquisition order supplies nested prefixes of 0, 1, 3, 7, 15, and 31 labels. At each budget the predictor conditions on the allowed prefix and predicts the same evaluation items.

Evaluation outcomes are used only to score those predictions. We average Brier scores over query items, then average repeated appearances within each task and give tasks equal weight. Brier ALC combines the six budgets with weights $(0.1, 0.2, 0.2, 0.2, 0.2, 0.1)$. Paired intervals resample whole tasks with their matched predictions.

# 4 Methodology

## 4.1 Models

**Beta-Binomial.** Let $p$ be a task's accuracy, with a Beta prior $p \sim \mathrm{Beta}(\alpha, \beta)$. After $n$ revealed labels, $k$ of them correct, the posterior is $\mathrm{Beta}(\alpha + k, \beta + n - k)$ and the prediction, the same for every item of the task, is its mean

$$\hat{p} = \frac{\alpha + k}{\alpha + \beta + n}. \tag{1}$$

The prior acts like $\alpha + \beta$ extra labels, $\alpha$ of them correct. The Jeffreys prior is $\alpha = \beta = \frac{1}{2}$ [9]. This is the exact Bayesian sequential update; nothing is learned.

**Rasch.** The probability that subject $i$ answers item $j$ correctly is

$$p_{ij} = \sigma(\theta_i - b_j), \qquad b_j = \delta_j - \mu_{d(j)}, \tag{2}$$

where $\sigma$ is the logistic function, $\theta_i$ is the subject's ability, and $b_j$ is the item's difficulty. This is the Rasch model [10] with the item difficulty split into a benchmark part and an item part: $\delta_j$ is the item's deviation from the average difficulty $-\mu_d$ of its benchmark $d = d(j)$, so a larger benchmark rate $\mu_d$ means an easier benchmark. The priors are $\mu_d \sim \mathcal{N}(m_0, \sigma_\mu^2)$, $\theta_i \sim \mathcal{N}(0, \sigma_\theta^2)$ and $\delta_j \sim \mathcal{N}(0, \sigma_\delta^2)$. At every budget we refit the maximum a posteriori (MAP) estimate from scratch on all labels revealed so far in the draw; an unlabelled item gets $\delta_j = 0$.

**Kernel Rasch.** Kernel Rasch keeps Equation (2) and changes only the prior on the item deviations $\boldsymbol{\delta} = (\delta_1, \ldots, \delta_M)^\top$. Rasch treats them as independent Gaussians, $\delta_j \sim \mathcal{N}(0, \sigma_\delta^2)$; kernel Rasch makes them jointly Gaussian, with covariance given by the similarity of the item embeddings:

$$\boldsymbol{\delta} \sim \mathcal{N}\big(\mathbf{0},\ K\big), \qquad K = \sigma_\delta^2\big[(1-\rho)\, I + \rho\, S\big], \quad S_{jj'} = \exp\Big(\frac{\cos(x_j, x_{j'}) - 1}{T}\Big), \tag{3}$$

where $x_j$ are the item embeddings of Section 3, $\rho \in [0, 1]$ weights the similarity part $S$ and $T$ is its temperature. At $\rho = 0$ the two priors coincide; for $\rho > 0$, similar items share difficulty, while items of different benchmarks stay independent. After the refit, a labelled item keeps its fitted deviation, and an unlabelled item $j$ receives the posterior mean given the labelled items $L$ of its benchmark, $\hat{\delta}_j = \rho\, \sigma_\delta^2\, S_{jL} K_{LL}^{-1} \hat{\boldsymbol{\delta}}_L$, which is 0 at $\rho = 0$ as in Rasch. With one subject per benchmark, as in the competition-shaped draws, no labels are shared across subjects: the joint fit equals independent per-task fits (maximum difference $10^{-14}$ in our check), so Rasch there differs from the Beta-Binomial only in its prior.

**Offline model that keeps its fitted values.** For the ID evaluation, the offline Rasch keeps its fitted $\mu_d$ and $\delta_j$ for a benchmark and items it has seen and updates only the subject ability $\theta_i$ online (a one-dimensional MAP on the task's own labels). For an unseen benchmark or item it falls back to $m_0$ and 0, i.e. to the fitted prior alone.

## 4.2 Probes

A probe predicts an item's relative difficulty from its embedding before any response from that benchmark is available. Our original controls use ridge regression on embeddings or log character count, and a sampled random score. The strengthened probes fit ridge regression and a one-hidden-layer MLP (32 units) to either item pass rate or negative fitted Rasch difficulty (easiness), with each target centered within benchmark. They use the same deterministic sample of 8,462 items, at most 800 per benchmark, from the 11 training benchmarks only. Inputs are standardized using fitting items only; regularization is selected in five folds grouped by training benchmark. The MLP has a 100-epoch limit. The Rasch targets come from the original offline fit, not a provisional refit.

We score a probe on each benchmark $d$ by two numbers, and average each over benchmarks. Let $y_j$ be item $j$'s observed pass rate across recorded subjects and $\hat{y}_j$ its predicted easiness; higher values mean easier items in both. The *Spearman correlation* $\rho_d = \mathrm{corr}\big(\mathrm{rank}(\hat{y}_j), \mathrm{rank}(y_j)\big)$ compares the two orderings of the items: it is 1 if the probe orders them perfectly, 0 if its order is unrelated to the truth, and $-1$ if it is reversed. The *pairwise accuracy* $A_d = \frac{1}{|P_d|} \sum_{(j,k) \in P_d} s_{jk}$ is the share

of item pairs the probe orders correctly, where $P_d$ is the set of pairs with $y_j \neq y_k$ (observed ties are excluded), $s_{jk} = 1$ if $\hat{y}_j - \hat{y}_k$ has the same sign as $y_j - y_k$, $s_{jk} = \frac{1}{2}$ if the probe predicts a tie, and $s_{jk} = 0$ otherwise. A random score has $A_d = 0.5$ on average. The ID diagnostic holds out items within the 11 training benchmarks; the OOD columns evaluate entire unseen benchmarks. All methods use matched items and sampled pairs.

To test whether a probe helps prediction, we give the item deviations of Rasch a prior mean taken from it,

$$\delta_j \sim \mathcal{N}\big(\gamma\, z_j,\ \sigma_\delta^2\big), \tag{4}$$

where $z_j$ is the probe's predicted difficulty of item $j$, centred within its benchmark (the benchmark rate $\mu_d$ absorbs the average), and $\gamma$ is a scale; $\gamma = 0$ is the fitted Rasch. For the Rasch-target arm, difficulty is the negative of the predicted easiness. For the pass-rate arm, the centered prediction $f(x_j)$ is converted to logit-scale difficulty as $\mathrm{logit}(\bar{p}) - \mathrm{logit}(\mathrm{clip}(\bar{p} + f(x_j), 10^{-4}, 1 - 10^{-4}))$, where $\bar{p}$ is the benchmark-weighted mean pass rate of the sampled training items. Positive difficulty means harder in both exports. At prediction time the probes use only item features; training outcomes supply supervision, while validation and test outcomes are used only for evaluation and validation-based selection of $\gamma$.

### 4.3 Priors

Each model has prior parameters: $(\alpha, \beta)$ for the Beta-Binomial; $m_0$ and the widths $\sigma_\mu, \sigma_\theta, \sigma_\delta$ for Rasch; and in addition $\rho$ and $T$ for kernel Rasch. Each is set from one of three sources.

- **Fixed in advance:** nothing learned. Beta-Binomial Jeffreys; Rasch $m_0 = 0$ and all widths 1; kernel Rasch additionally $\rho = 0.5$, $T = 0.1$.
- **Tuned on validation:** the best setting on the validation benchmarks from a grid (Beta-Binomial: $\alpha = 0.4$, $\beta = 0.6$; Rasch: $m_0 = 0.048$, $\sigma_\mu = 2$, $\sigma_\theta = 0.5$, $\sigma_\delta = 1$; kernel Rasch: $m_0 = 0.048$, $\sigma_\mu = 2$, $\sigma_\theta = 1$, $\sigma_\delta = 1.4$, $\rho = 0.8$, $T = 0.2$; probe prior mean, Section 4.2: the scale $\gamma$ from a grid).
- **Fitted on training:** the 11 training benchmarks only. The Beta-Binomial prior maximises the Beta-Binomial marginal likelihood of the training tasks, weighting benchmarks equally ($\alpha = 1.92$, $\beta = 1.76$: a prior mean of $0.522$ worth $\alpha + \beta = 3.68$ labels). The Rasch priors come from a joint MAP fit to $4.34$ million training responses (4,809 subjects, 109,987 items) with empirical-Bayes hyperparameters averaged per benchmark so that the benchmark with most subjects does not dominate: $m_0 = 0.175$, $\sigma_\mu = 0.75$, $\sigma_\theta = 1.38$, $\sigma_\delta = 1.49$. Kernel Rasch uses the same values and keeps $\rho = 0.8$ and $T = 0.2$ from the validation tuning, so its kernel is not fitted on training. The probes of Section 4.2 are trained on the training benchmarks' items only. Only these numbers transfer; the fitted $\mu_d$, $\theta_i$ and $\delta_j$ belong to training benchmarks, subjects and items.

## 5 Results

Every difference is written $\Delta_{a-b}$, the Brier ALC of model $a$ minus that of model $b$, so negative favours $a$; paired 95% bootstrap intervals are over tasks. J is the Jeffreys Beta-Binomial; BB, R and KR are the Beta-Binomial, Rasch and kernel Rasch models, subscripted fix, tune or fit for the source of their prior, and R off is the offline Rasch that keeps its fitted values.

### 5.1 Prior source matters, and no source wins on every split

Table 1 compares the nine combinations. On test, a Beta-Binomial with a prior fitted on training is the best predictor ($\Delta_{\mathrm{BB\,fit-J}} = -0.0138$ [$-0.0158$, $-0.0116$]), four times the gain of the validation-tuned Rasch, and better than any Rasch model ($\Delta_{\mathrm{R\,fit-BB\,fit}} = +0.0019$ [$+0.0007$, $+0.0031$]). The validation-tuned Beta-Binomial prior is *worse* than Jeffreys on test ($\Delta_{\mathrm{BB\,tune-J}} = +0.0055$) though it is the best Beta-Binomial on validation. Every comparison between prior sources changes sign between test and validation (Appendix B); fresh draws change no result by more than $0.0007$ (Appendix C).

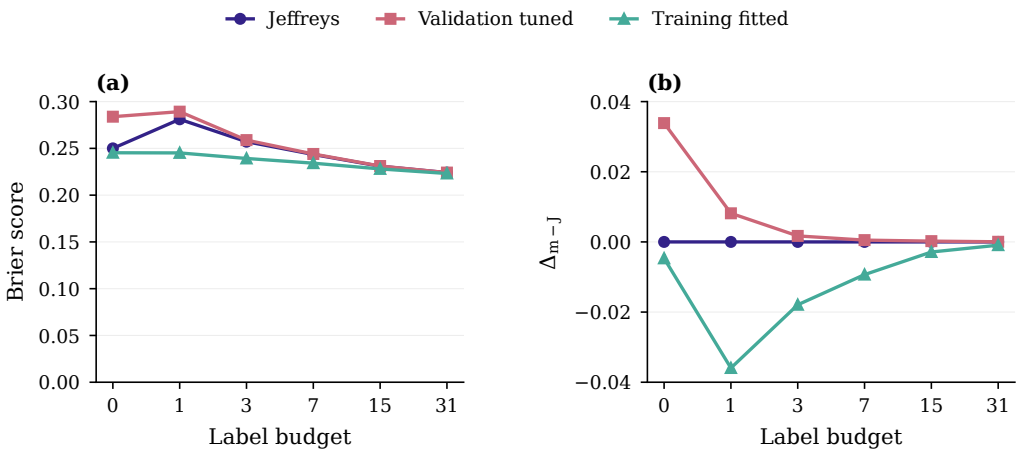



Figure 1: **The prior's source matters at zero to three labels and has faded by 31.** Brier score (a) and $\Delta_{m-\mathrm{J}}$ (b) per label budget on the three unseen test benchmarks, competition-shaped draws, for the Beta-Binomial with a fixed (Jeffreys, solid), validation-tuned (dashed) and training-fitted (dotted) prior. Jeffreys jumps to $0.75$ or $0.25$ after one answer, so the gap is largest at one label; at 31 labels the three differ by less than $0.001$.

Table 1: **A prior fitted on training helps on the test benchmarks and hurts on the validation benchmarks; a fixed prior with strong shrinkage does nearly as well on test.** Each cell is $\Delta_{m-\mathrm{J}}$ for the model $m$ in the row, competition-shaped draws. ID: held-out subjects on the 11 training benchmarks, 114 tasks, where the training-fitted Beta-Binomial prior is refit without those subjects. OOD validation: 3 unseen benchmarks, 80 tasks. OOD test: 3 unseen benchmarks, 45 tasks. Bold: best of the nine models in each column; underlined: second best.

| Model $m$ | ID | OOD validation | OOD test |
|---|---|---|---|
| *Fixed in advance* | | | |
| Beta-Binomial (Jeffreys) | 0 (reference) | 0 (reference) | 0 (reference) |
| Rasch | $-0.0073$ $[-0.0091, -0.0055]$ | $+0.0031$ $[-0.0005, +0.0066]$ | $-0.0102$ $[-0.0118, -0.0085]$ |
| Kernel Rasch | $-0.0072$ $[-0.0087, -0.0058]$ | $+0.0029$ $[-0.0008, +0.0065]$ | $-0.0105$ $[-0.0121, -0.0088]$ |
| *Tuned on validation* | | | |
| Beta-Binomial | $+0.0041$ $[+0.0027, +0.0055]$ | $\mathbf{-0.0015}$ $[-0.0040, +0.0010]$ | $+0.0055$ $[+0.0038, +0.0072]$ |
| Rasch | $-0.0024$ $[-0.0028, -0.0019]$ | $+0.0004$ $[-0.0006, +0.0014]$ | $-0.0032$ $[-0.0035, -0.0029]$ |
| Kernel Rasch | $+0.0001$ $[-0.0004, +0.0005]$ | $\underline{-0.0003}$ $[-0.0010, +0.0004]$ | $-0.0024$ $[-0.0027, -0.0021]$ |
| *Fitted on training* | | | |
| Beta-Binomial | $\mathbf{-0.0090}$ $[-0.0115, -0.0063]$ | $+0.0079$ $[+0.0024, +0.0136]$ | $\mathbf{-0.0138}$ $[-0.0158, -0.0116]$ |
| Rasch | $\underline{-0.0081}$ $[-0.0101, -0.0060]$ | $+0.0052$ $[+0.0008, +0.0097]$ | $\underline{-0.0119}$ $[-0.0133, -0.0106]$ |
| Kernel Rasch | $-0.0063$ $[-0.0076, -0.0049]$ | $+0.0036$ $[-0.0007, +0.0076]$ | $-0.0118$ $[-0.0131, -0.0105]$ |

### 5.2 Learned priors transfer to unseen benchmarks; item-level estimates do not

On the ID evaluation set the offline model that keeps its fitted values scores $\Delta_{\mathrm{R\,off}-\mathrm{J}} = -0.0852$ $[-0.0955, -0.0747]$ against $\Delta_{\mathrm{BB\,fit}-\mathrm{J}} = -0.0090$ for the fitted Beta-Binomial prior and $\Delta_{\mathrm{R\,fit}-\mathrm{J}} = -0.0081$ for the fitted Rasch prior (Figure 2); its item difficulties alone, at budget zero, reach a within-task AUC of $0.836$. On the unseen test benchmarks it equals the fitted prior ($\Delta_{\mathrm{R\,off}-\mathrm{J}} = -0.0115$ against $\Delta_{\mathrm{R\,fit}-\mathrm{J}} = -0.0119$): nothing it knows about particular items or benchmark rates is available. This is the seen-benchmark check the design requires, and it separates two things that are easy to conflate: item-level estimates transfer only to the items they were fitted on, while the prior's spreads carry over to new benchmarks.

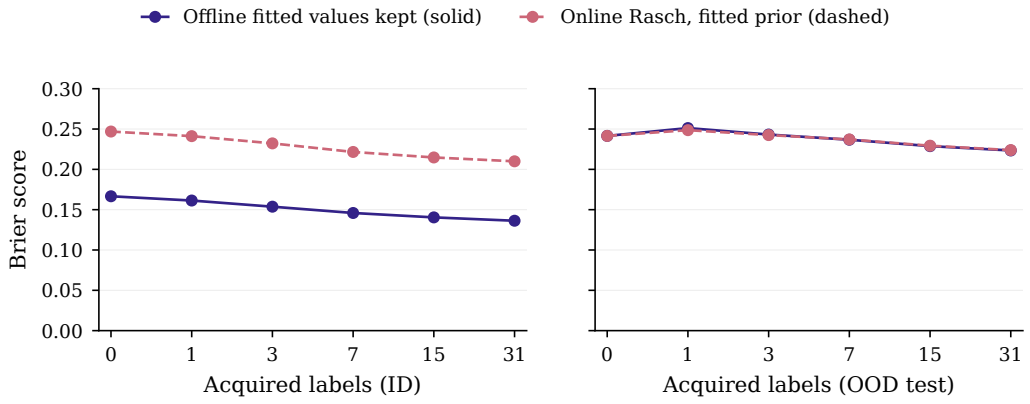



Figure 2: **Offline estimates win on familiar benchmarks and have no edge on unfamiliar ones.** Brier score per label budget for the offline Rasch keeping its fitted benchmark rates and item difficulties and updating only ability (solid), against the online Rasch with the same fitted prior (dashed). Left: ID, held-out subjects on the 11 training benchmarks. Right: OOD test benchmarks, where the offline model has no fitted values and falls back to the prior.

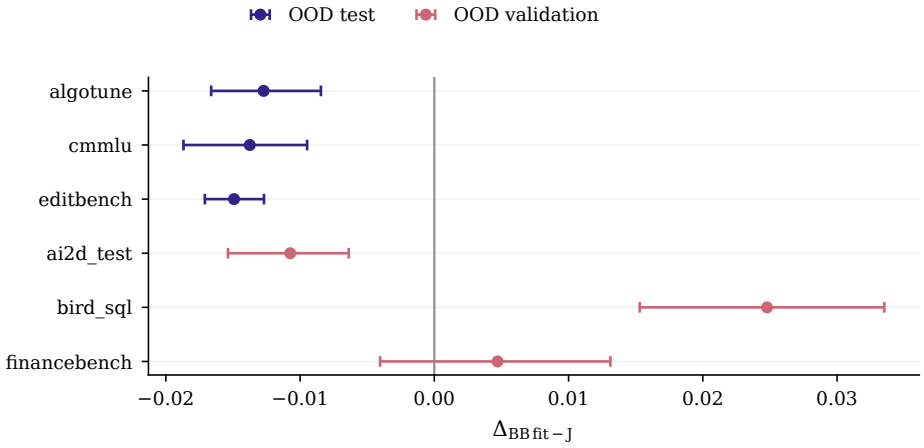



Figure 3: **The training-fitted prior helps on all three test benchmarks and on one validation benchmark, and hurts on the hardest.** $\Delta_{\mathrm{BB\,fit-J}}$ for each unseen benchmark, with 95% intervals over tasks. The validation benchmark `bird_sql` has task accuracy 0.12, far below every training benchmark.

### 5.3 Learned priors help benchmarks of typical difficulty and hurt much easier or harder ones

Figure 3 shows the test and validation benchmarks separately: all three test benchmarks agree ($\Delta_{\mathrm{BB\,fit-J}}$ from $-0.013$ to $-0.015$), and on validation `ai2d_test` agrees while `bird_sql` goes the other way ($+0.025$). Over the 17 benchmarks (Figure 4; each training benchmark is held out in turn, Appendix D) the fitted prior significantly helps on 9, significantly hurts on 1 (`bird_sql`), and the macro mean $\Delta_{\mathrm{BB\,fit-J}}$ is $-0.0050$. What separates the benchmarks is not whether they lie inside the range of training benchmark rates (`alignment_faking` and `harmbench` lose while inside it) but how far their task accuracies sit from the prior mean: the Spearman correlation between the gain and that distance is 0.96 over the 17 benchmarks (0.93 over the 11 training benchmarks), against 0.01 over the 11 for the benchmark's mean accuracy. The prior is a starting guess that a task is answered correctly about 52% of the time, and with few labels the predictor pulls its estimate toward it. If a benchmark's true accuracy is, say, 55%, the pull lands near the truth and the prediction improves; if

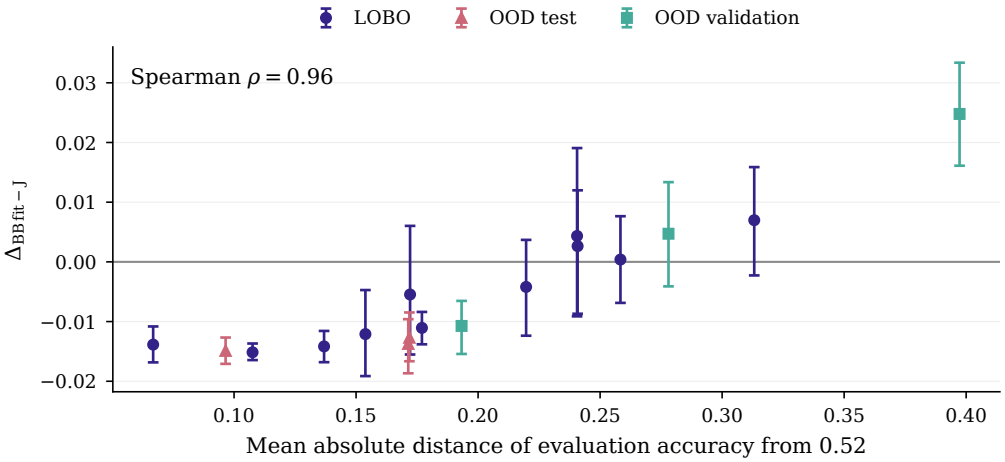



Figure 4: **A fitted prior helps when a benchmark's accuracy is near the prior mean.** Each point is one of 17 benchmarks. For each of the 11 training benchmarks, the prior was refit without any of that benchmark's rows and scored on held-out subjects. The six validation and test benchmarks use the original training-only prior. The horizontal axis uses task accuracy on the evaluation items; the vertical axis shows $\Delta_{\text{BB fit}-\text{J}}$, with 95% task-bootstrap intervals. This is an explanatory analysis using observed outcomes.

it is 12%, as on `bird_sql`, the pull drags the estimate the wrong way and it gets worse. So the prior helps on benchmarks of typical difficulty and hurts on much easier or much harder ones. The three test benchmarks (mean accuracy 0.54–0.68) sit near the prior mean, which is why test looks good. We found this pattern after seeing the results, so it is an explanation and not a predictor.

Under the fitted priors, the simplest Rasch model is also the best on test and the worst on validation: with benchmark rate only $\Delta_{m-\text{J}}$ is $-0.0172$ on test and $+0.0211$ on validation (Appendix A). The subject-ability term is what lets a benchmark far from the prior move away from it, at a cost in variance when the benchmark is near it. The kernel adds nothing on test.

### 5.4 Item content carries a small rank signal and a negligible gain for the prior

Table 2: Item-difficulty transfer: within-benchmark Spearman correlation and pairwise accuracy between predicted easiness and observed pass rate, averaged across benchmarks. ID: items held out within the 11 training benchmarks; OOD validation and OOD test: three unseen benchmarks each. The additional pass-rate and Rasch-target probes use benchmark-grouped selection and were not evaluated on ID (–).

| | ID | | OOD validation | | OOD test | |
|---|---|---|---|---|---|---|
| Probe | Spearman | Pairwise | Spearman | Pairwise | Spearman | Pairwise |
| Random score | −0.003 | 0.499 | −0.024 | 0.490 | 0.076 | 0.529 |
| Log character count | 0.107 | 0.539 | 0.058 | 0.525 | 0.061 | 0.521 |
| Ridge on embeddings | 0.400 | 0.649 | 0.196 | 0.570 | −0.020 | 0.494 |
| Pass-rate ridge | – | – | 0.165 | 0.557 | −0.035 | 0.487 |
| Rasch-target ridge | – | – | 0.130 | 0.544 | −0.035 | 0.487 |
| Pass-rate MLP | – | – | 0.071 | 0.523 | 0.109 | 0.539 |
| Rasch-target MLP | – | – | 0.146 | 0.550 | 0.059 | 0.522 |

Embeddings show a clear signal on known benchmarks. On the six unseen benchmarks the two MLP probes pass the criterion we fixed in advance, with benchmark-macro Spearman 0.090 and 0.103 against 0.026 for a random baseline and a positive correlation on 5 of 6 benchmarks; the ridge probes do not (0.065 and 0.047). On the three test benchmarks alone, the pass-rate MLP reaches

$0.109$, while the Rasch-target MLP reaches $0.059$, below the sampled random control's $0.076$. The aggregate rank criterion therefore does not establish a consistent test-only advantage.

The rank signal yields only a small gain at the validation-selected scale. In Table 3, every $\gamma \geq 0.25$ is worse than $\gamma = 0$ on both splits. The scale selected on validation ($\gamma = 0.05$, Rasch-target MLP) improves Brier ALC by about $0.0004$ on each split in its matched replay; this comparison alone does not establish a reliable improvement. This tests these two probes only; a stronger difficulty predictor, for instance one trained through an item response likelihood on more items, could do better.

Table 3: Predicted item difficulty as the prior mean of the Rasch item deviation, Equation (4). Cells are $\Delta_{m-\mathrm{J}}$ of the fitted Rasch with scale $\gamma$, competition-shaped draws; $\gamma = 0$ is the fitted Rasch without a prior mean. Bold: lowest in each column.

| | Pass-rate MLP | | Rasch-target MLP | |
|---|---|---|---|---|
| Scale $\gamma$ | OOD validation | OOD test | OOD validation | OOD test |
| 0 | $\mathbf{+0.0052}$ | $-0.0119$ | $+0.0052$ | $-0.0119$ |
| 0.05 | $+0.0062$ | $\mathbf{-0.0119}$ | $\mathbf{+0.0048}$ | $\mathbf{-0.0123}$ |
| 0.1 | $+0.0103$ | $-0.0082$ | $+0.0051$ | $-0.0120$ |
| 0.25 | $+0.0332$ | $+0.0150$ | $+0.0099$ | $-0.0079$ |
| 0.5 | $+0.0690$ | $+0.0491$ | $+0.0273$ | $+0.0069$ |
| 1 | $+0.1065$ | $+0.0826$ | $+0.0704$ | $+0.0433$ |
| 1.5 | $+0.1326$ | $+0.1050$ | $+0.1040$ | $+0.0724$ |

## 6 Limitations

The filtered public data covers only three test benchmarks in the online simulation. Multiple draws and acquisition orders measure variation within this population; they do not supply independent tests of transfer to new benchmarks. Task bootstrap intervals likewise do not capture uncertainty from sampling new benchmarks. Results apply to the eligible response data and random label acquisition, and need not carry over to other benchmark mixes or selection policies. The 17-benchmark sensitivity analysis reuses held-out-subject tasks for the 11 original training benchmarks. The test triples overlap, so their distribution describes sensitivity within this dataset rather than independent estimates of transfer. The ID comparison has item response history that is unavailable under benchmark cold start. Observed item difficulty also depends on which subjects answered the item. The leave-one-benchmark-out association uses realized outcomes to explain when the fitted prior helps; it is not a way to select a prior before observing a new benchmark.

Further limits of our own. Each competition-shaped draw has one subject per benchmark, so sharing labels across subjects is not tested. The test gain depends on this split. The test benchmarks are close to the prior mean; over all 680 choices of three held-out benchmarks the fitted prior gains less ($\Delta_{\mathrm{BB\,fit}-\mathrm{J}} = -0.0037$ on average, refit on the other 14) and hurts in $17.5\%$ of them (Appendix E), as it does on validation. Test scores were inspected repeatedly while building the models, so the test analyses are exploratory: all selection used validation, but test was not read once. The kernel settings were chosen on validation and not on the ID evaluation set. We did not rerun the subject-metadata prior in this design, and we dropped reasoning-trace features, which are not available at test time.

## 7 Conclusion

What transfers from offline fitting to an unseen benchmark is a shrinkage prior, not item-level estimates. Where fitted values apply, offline fitting is worth a very large gain ($\Delta_{\mathrm{R\,off}-\mathrm{J}} = -0.0852$ on the ID evaluation set); where they do not, it is worth none. The prior's gain is modest and depends on the benchmark: $-0.0050$ on average over the 17 benchmarks and $-0.0037$ over the 680 choices of three held-out benchmarks, with a loss in $17.5\%$ of them. The prior helps when a benchmark's accuracy lies near the prior mean and hurts when it lies far away, which is why the same fitted prior looked good on test and bad on validation. A well-fitted Beta-Binomial is the strongest estimator in our test, though not on validation; whether sharing labels across subjects helps remains untested, since each draw has one subject per benchmark. Two attempts to do better did not help: priors that adapt to the first labels did not remove the harm without losing the gain, and the neural probes we

trained on item embeddings, which carry a small rank signal for difficulty, improved prediction by only about $0.0004$ Brier ALC as a prior mean.

## Disclosure and rerun instructions

**Training data.** All fitted quantities (the offline Rasch fit, the Beta-Binomial empirical-Bayes prior, the item probes) use only the responses of the 11 training benchmarks in the competition's measurement database after the filtering described in Section 3; no other response data is used. **Pretrained models.** Item text is embedded with the frozen BGE-M3 model; nothing else is pretrained or fine-tuned. **LLM coding assistants.** Claude Code (Anthropic) assisted with writing code, analysis scripts and drafting text; every reported number was produced by running the code in the repository, and the authors checked the numbers. OpenAI Codex assisted with the reviewer analyses, figures, and text revisions.

The shared handoff contains frozen draws, scores, and competition-shaped test predictions. The GitHub analysis branch contains online-simulation and analysis code, aggregate results, and figure PDFs. Keep private response and prediction files in the authorized shared storage. The command `python organized/testing/eandd/wp9.py` with `-handoff` pointing to the local handoff and `-output` pointing to a fresh directory rebuilds the figures and checks the saved Brier scores against the saved predictions. The analysis README identifies the handoff location and documents how to render figures from aggregate CSVs alone. The original item-probe results are in `paper/eandd/data/wp7-summary.csv`; the confirmed neural-probe results are in `paper/eandd/data/wp13-probe-summary.csv`. Record the analysis commit and retain the frozen draws for any rerun.

## References


[1] H. Robbins. An empirical Bayes approach to statistics. In *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability*, 1:157–163, 1956.

[2] B. Efron and C. Morris. Stein's estimation rule and its competitors—an empirical Bayes approach. *Journal of the American Statistical Association*, 68(341):117–130, 1973.

[3] J. N. K. Rao and I. Molina. *Small Area Estimation*. Wiley, 2nd edition, 2015.

[4] A. N. Angelopoulos, S. Bates, C. Fannjiang, M. I. Jordan, and T. Zrnic. Prediction-powered inference. *Science*, 382(6671):669–674, 2023.

[5] M. Jeon, I. H. Jin, M. Schweinberger, and S. Baugh. Mapping unobserved item–respondent interactions: A latent space item response model with interaction map. *Psychometrika*, 86(2):378–403, 2021.

[6] G. Lior, T. Frostig, G. Stanovsky, and M. Eyal. Extending item response theory for efficient and meaningful multilingual evaluation. arXiv:2606.15643, 2026.

[7] G. W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1950.

[8] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, Z. Liu. BGE M3-Embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation. arXiv:2402.03216, 2024.

[9] H. Jeffreys. An invariant form for the prior probability in estimation problems. *Proc. Royal Society A*, 186, 1946.

[10] G. Rasch. *Probabilistic Models for Some Intelligence and Attainment Tests*. Danish Institute for Educational Research, 1960.

[11] S. T. Truong, Y. Tu, P. Liang, B. Li, S. Koyejo. Reliable and efficient amortized model-based evaluation. In *ICML*, PMLR 267, 2025.

[12] G. Zhang, F. E. Dorner, M. Hardt. How benchmark prediction from fewer data misses the mark. arXiv:2506.07673, 2025.

# A Rasch terms under the fitted priors

Table 4: $\Delta_{m-\mathrm{J}}$ for Rasch with terms switched off (prior width $\sigma$ set near zero), under the training-fitted priors. Competition-shaped draws on OOD validation and OOD test. Bold: lowest in each column.

| Model $m$ | OOD validation | OOD test |
|---|---|---|
| Benchmark rate $\mu_d$ only | $+0.0211$<br>$[+0.0114, +0.0311]$ | $-0.0172$<br>$[-0.0202, -0.0142]$ |
| + subject ability $\theta_i$ | $\mathbf{+0.0029}$<br>$[-0.0003, +0.0060]$ | $-0.0089$<br>$[-0.0100, -0.0080]$ |
| + item deviation $\delta_j$ (Rasch) | $+0.0052$<br>$[+0.0007, +0.0098]$ | $-0.0119$<br>$[-0.0133, -0.0106]$ |
| + kernel | $+0.0036$<br>$[-0.0006, +0.0078]$ | $-0.0118$<br>$[-0.0131, -0.0105]$ |
| $\mu_d$ and $\delta_j$, no $\theta_i$ | $+0.0282$<br>$[+0.0164, +0.0401]$ | $\mathbf{-0.0174}$<br>$[-0.0208, -0.0139]$ |
| $\theta_i$ and $\delta_j$, no $\mu_d$ | $+0.0078$<br>$[+0.0022, +0.0135]$ | $-0.0140$<br>$[-0.0157, -0.0122]$ |

# B Paired comparisons between prior sources

Table 5: Paired $\Delta_{a-b}$ between prior sources, with 95% intervals over tasks. Each comparison changes sign between OOD test and OOD validation. A dash: not run on ID.

| Comparison | ID | OOD validation | OOD test |
|---|---|---|---|
| $\Delta_{\mathrm{R\,fit}-\mathrm{R\,fix}}$ | – | $+0.0021$<br>$[+0.0006, +0.0036]$ | $-0.0017$<br>$[-0.0028, -0.0005]$ |
| $\Delta_{\mathrm{KR\,fit}-\mathrm{R\,fit}}$ | – | $-0.0016$<br>$[-0.0021, -0.0012]$ | $+0.0001$<br>$[-0.0002, +0.0004]$ |
| $\Delta_{\mathrm{R\,fit}-\mathrm{BB\,fit}}$ | $+0.0009$<br>$[-0.0001, +0.0018]$ | $-0.0027$<br>$[-0.0041, -0.0012]$ | $+0.0019$<br>$[+0.0007, +0.0031]$ |

# C Draw-seed robustness

Table 6: $\Delta_{m-\mathrm{J}}$ on three fresh sets of 150 competition-shaped draws per role (seeds 20261201–03); results on the frozen draws are in Table 1.

| Model $m$ | OOD validation | OOD test |
|---|---|---|
| BB fit | $+0.0078$ / $+0.0083$ / $+0.0089$ | $-0.0137$ / $-0.0144$ / $-0.0139$ |
| R fit | $+0.0051$ / $+0.0054$ / $+0.0060$ | $-0.0120$ / $-0.0124$ / $-0.0123$ |
| BB tune | $-0.0015$ / $-0.0016$ / $-0.0018$ | $+0.0060$ / $+0.0059$ / $+0.0063$ |

## D Leave-one-benchmark-out for the fitted prior

Table 7: $\Delta_{\mathrm{BB\,fit-J}}$ with the prior refit without the held-out benchmark, on each training benchmark's held-out-subject tasks, with the mean distance of its task accuracies from the prior mean $0.52$.

| Held-out benchmark | Tasks | $\Delta_{\mathrm{BB\,fit-J}}$ [95% CI] | Mean distance from $0.52$ |
|---|---|---|---|
| `hallusionbench` | 20 | $-0.0151$ [$-0.0165$, $-0.0136$] | 0.11 |
| `bridging_gap` | 20 | $-0.0142$ [$-0.0167$, $-0.0114$] | 0.14 |
| `helm_afr` | 3 | $-0.0139$ [$-0.0168$, $-0.0108$] | 0.07 |
| `afrimedqa` | 6 | $-0.0121$ [$-0.0189$, $-0.0046$] | 0.15 |
| `afrieval` | 20 | $-0.0111$ [$-0.0138$, $-0.0085$] | 0.18 |
| `cruxeval` | 5 | $-0.0055$ [$-0.0155$, $+0.0056$] | 0.17 |
| `brace` | 6 | $-0.0042$ [$-0.0121$, $+0.0037$] | 0.22 |
| `agi_elo` | 4 | $+0.0004$ [$-0.0069$, $+0.0076$] | 0.26 |
| `appworld` | 4 | $+0.0026$ [$-0.0087$, $+0.0120$] | 0.24 |
| `alignment_faking` | 20 | $+0.0043$ [$-0.0092$, $+0.0188$] | 0.24 |
| `harmbench` | 6 | $+0.0070$ [$-0.0023$, $+0.0161$] | 0.31 |
| Mean over the 11 benchmarks | 114 | $-0.0056$ | 0.19 |

## E Sensitivity to the benchmark split

We held out each of the 680 triples, refitted the benchmark-weighted empirical Beta prior on the other 14 benchmarks, and rescored the frozen trajectories. The mean macro $\Delta_{\mathrm{BB\,fit-J}}$ was $-0.0037$, with 5th–95th percentiles [$-0.0095$, $+0.0099$]. The fitted prior hurt on 17.5% of splits. The original test benchmark names gave $-0.0087$ (percentile 13) under refitting. This refit includes the former validation benchmarks among the 14 eligible fit benchmarks for the original test triple, so its result differs from the original 11-benchmark fit. The 11 original training benchmarks are scored on held-out-subject tasks from the existing leave-one-out data. These overlapping triples describe split sensitivity within this dataset.

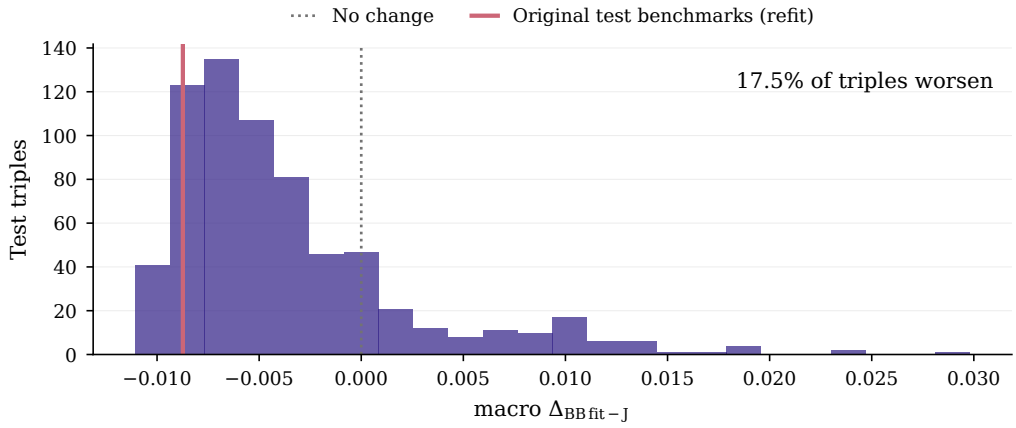



Figure 5: $\Delta_{\mathrm{BB\,fit-J}}$ over all 680 triples. Each empirical prior is fitted on the 14 benchmarks outside its triple. The vertical line marks the original test benchmark names under this refitting protocol.

# F Alternative evaluation metrics

Table 8: Competition-shaped online simulation; models are labelled as in Section 5. Brier, log loss, and ten-bin task ECE are averaged using the official budget weights; lower is better. AUC uses within-task item rankings at 31 labels, where higher is better. Dashes indicate no informative ranking; a constant prediction ties all items. Bold: best in each split (lowest loss, highest AUC). Per-budget AUC and alternative weights are reported in the accompanying CSVs.

| Split | Model $m$ | Brier | Log loss | ECE | AUC at 31 labels |
|---|---|---|---|---|---|
| ID | J | 0.2357 | 0.6681 | 0.1407 | – |
| | BB fit | 0.2268 | 0.6437 | 0.1235 | – |
| | R fit | 0.2276 | 0.6460 | 0.1245 | – |
| | KR fit | 0.2295 | 0.6506 | 0.1336 | 0.581 |
| | R off | **0.1506** | **0.4634** | **0.1161** | **0.836** |
| OOD validation | J | **0.1738** | **0.5159** | **0.1405** | – |
| | BB fit | 0.1817 | 0.5404 | 0.1677 | – |
| | R fit | 0.1790 | 0.5323 | 0.1580 | – |
| | KR fit | 0.1774 | 0.5279 | 0.1603 | **0.618** |
| OOD test | J | 0.2499 | 0.7045 | 0.1426 | – |
| | BB fit | **0.2362** | **0.6656** | **0.1113** | – |
| | R fit | 0.2380 | 0.6712 | 0.1164 | – |
| | KR fit | 0.2382 | 0.6717 | 0.1230 | **0.571** |